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CSES Problem Set**Meet in the Middle**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n numbers. In how many ways can you choose a subset of the numbers with sum x ?

Input

The first input line has two numbers n and x : the array size and the required sum.

The second line has n integers $t_1, t_2, \dots, t_n$: the numbers in the array.

Output

Print the number of ways you can create the sum x .

Constraints

- $1 \leq n \leq 40$
- $1 \leq x \leq 10^9$
- $1 \leq t_i \leq 10^9$

Example

Input:

4 5
1 2 3 2

Output:

3

CSES Problem Set

Hamming Distance

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

The Hamming distance between two strings a and b of equal length is the number of positions where the strings differ.

You are given n bit strings, each of length k and your task is to calculate the minimum Hamming distance between two strings.

Input

The first input line has two integers n and k : the number of bit strings and their length.

Then there are n lines each consisting of one bit string of length k .

Output

Print the minimum Hamming distance between two strings.

Constraints

- $2 \leq n \leq 2 \cdot 10^4$
- $1 \leq k \leq 30$

Example

Input:

```
5 6
110111
001000
100001
101000
101110
```

Output:

```
1
```

Explanation: The strings 101000 and 001000 differ only at the first position.

CSES Problem Set

Corner Subgrid Count

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an $n \times n$ grid whose each square is either black or white. A subgrid is called *beautiful* if its height and width is at least two and all of its corners are black. How many beautiful subgrids are there within the given grid?

Input

The first input line has an integer n : the size of the grid.

Then there are n lines describing the grid: 1 means that a square is black and 0 means it is white.

Output

Print the number of beautiful subgrids.

Constraints

- $1 \leq n \leq 3000$

Example

Input:

```
5
00010
11111
00110
11001
00010
```

Output:

```
4
```

CSES Problem Set

Reachable Nodes

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A directed acyclic graph consists of n nodes and m edges. The nodes are numbered $1, 2, \dots, n$.

Calculate for each node the number of nodes you can reach from that node (including the node itself).

Input

The first input line has two integers n and m : the number of nodes and edges.

Then there are m lines describing the edges. Each line has two distinct integers a and b : there is an edge from node a to node b .

Output

Print n integers: for each node the number of reachable nodes.

Constraints

- $1 \leq n \leq 5 \cdot 10^4$
- $1 \leq m \leq 10^5$

Example

Input:

```
5 6
1 2
1 3
1 4
2 3
3 5
4 5
```

Output:

```
5 3 2 2 1
```

CSES Problem Set

Reachability Queries

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A directed graph consists of n nodes and m edges. The edges are numbered $1, 2, \dots, n$.

Your task is to answer q queries of the form "can you reach node b from node a ?"

Input

The first input line has three integers n , m and q : the number of nodes, edges and queries.

Then there are m lines describing the edges. Each line has two distinct integers a and b : there is an edge from node a to node b .

Finally there are q lines describing the queries. Each line consists of two integers a and b : "can you reach node b from node a ?"

Output

Print the answer for each query: either "YES" or "NO".

Constraints

- $1 \leq n \leq 5 \cdot 10^4$
- $1 \leq m, q \leq 10^5$

Example

Input:

```
4 4 3
1 2
2 3
3 1
4 3
1 3
1 4
4 1
```

Output:

```
YES
NO
YES
```

CSES Problem Set

Cut and Paste

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a string, your task is to process operations where you cut a substring and paste it to the end of the string. What is the final string after all the operations?

Input

The first input line has two integers n and m : the length of the string and the number of operations. The characters of the string are numbered $1, 2, \dots, n$.

The next line has a string of length n that consists of characters A–Z.

Finally, there are m lines that describe the operations. Each line has two integers a and b : you cut a substring from position a to position b .

Output

Print the final string after all the operations.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a \leq b \leq n$

Example

Input:

```
7 2
AYBABTU
3 5
3 5
```

Output:

```
AYABTUB
```

CSES Problem Set

Substring Reversals

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a string, your task is to process operations where you reverse a substring of the string. What is the final string after all the operations?

Input

The first input line has two integers n and m : the length of the string and the number of operations. The characters of the string are numbered $1, 2, \dots, n$.

The next line has a string of length n that consists of characters A–Z.

Finally, there are m lines that describe the operations. Each line has two integers a and b : you reverse a substring from position a to position b .

Output

Print the final string after all the operations.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a \leq b \leq n$

Example

Input:

```
7 2
AYBABTU
3 4
4 7
```

Output:

```
AYAUTBB
```

CSES Problem Set

Reversals and Sums

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n integers, you have to process following operations:

1. reverse a subarray
2. calculate the sum of values in a subarray

Input

The first input line has two integers n and m : the size of the array and the number of operations. The array elements are numbered $1, 2, \dots, n$.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Finally, there are m lines that describe the operations. Each line has three integers t, a and b . If $t = 1$, you should reverse a subarray from a to b . If $t = 2$, you should calculate the sum of values from a to b .

Output

Print the answer to each operation where $t = 2$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 10^5$
- $0 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input:

```
8 3
2 1 3 4 5 3 4 4
2 2 4
1 3 6
2 2 4
```

Output:

```
8
9
```

CSES Problem Set

Necessary Roads

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m roads between them. There is a route between any two cities.

A road is called *necessary* if there is no route between some two cities after removing that road. Your task is to find all necessary roads.

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the roads. Each line has two integers a and b : there is a road between cities a and b . There is at most one road between two cities, and every road connects two distinct cities.

Output

First print an integer k : the number of necessary roads. After that, print k lines that describe the roads. You may print the roads in any order.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 5
1 2
1 4
2 4
3 5
4 5
```

Output:

```
2
3 5
4 5
```

CSES Problem Set

Necessary Cities

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m roads between them. There is a route between any two cities.

A city is called *necessary* if there is no route between some other two cities after removing that city (and adjacent roads). Your task is to find all necessary cities.

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the roads. Each line has two integers a and b : there is a road between cities a and b . There is at most one road between two cities, and every road connects two distinct cities.

Output

First print an integer k : the number of necessary cities. After that, print a list of k cities. You may print the cities in any order.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 5
1 2
1 4
2 4
3 5
4 5
```

Output:

```
2
4 5
```

CSES Problem Set**Eulerian Subgraphs**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an undirected graph that has n nodes and m edges.

We consider subgraphs that have all nodes of the original graph and some of its edges. A subgraph is called *Eulerian* if each node has even degree.

Your task is to count the number of Eulerian subgraphs modulo $10^9 + 7$.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the edges. Each line has two integers a and b : there is an edge between nodes a and b . There is at most one edge between two nodes, and each edge connects two distinct nodes.

Output

Print the number of Eulerian subgraphs modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
4 3
1 2
1 3
2 3
```

Output:

```
2
```

Explanation: You can either keep or remove all edges, so there are two possible Eulerian subgraphs.

CSES Problem Set

Monster Game I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are playing a game that consists of n levels. Each level has a monster. On levels $1, 2, \dots, n-1$, you can either kill or escape the monster. However, on level n you must kill the final monster to win the game.

Killing a monster takes sf time where s is the monster's strength and f is your skill factor (lower skill factor is better). After killing a monster, you get a new skill factor. What is the minimum total time in which you can win the game?

Input

The first input line has two integers n and x : the number of levels and your initial skill factor.

The second line has n integers $s_1, s_2, \dots, s_n$: each monster's strength.

The third line has n integers $f_1, f_2, \dots, f_n$: your new skill factor after killing a monster.

Output

Print one integer: the minimum total time to win the game.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$
- $1 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq 10^6$
- $x \geq f_1 \geq f_2 \geq \dots \geq f_n \geq 1$

Example

Input:

```
5 100
20 30 30 50 90
90 60 20 20 10
```

Output:

```
4800
```

Explanation: The best way to play is to kill the third and fifth monster.

CSES Problem Set

Monster Game II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are playing a game that consists of n levels. Each level has a monster. On levels $1, 2, \dots, n-1$, you can either kill or escape the monster. However, on level n you must kill the final monster to win the game.

Killing a monster takes sf time where s is the monster's strength and f is your skill factor. After killing a monster, you get a new skill factor (lower skill factor is better). What is the minimum total time in which you can win the game?

Input

The first input line has two integers n and x : the number of levels and your initial skill factor.

The second line has n integers $s_1, s_2, \dots, s_n$: each monster's strength.

The third line has n integers $f_1, f_2, \dots, f_n$: your new skill factor after killing a monster.

Output

Print one integer: the minimum total time to win the game.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$
- $1 \leq s_i, f_i \leq 10^6$

Example

Input:

```
5 100
50 20 30 90 30
60 20 20 10 90
```

Output:

```
2600
```

Explanation: The best way to play is to kill the second and fifth monster.

CSES Problem Set

Subarray Squares

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n elements, your task is to divide into k subarrays. The cost of each subarray is the square of the sum of the values in the subarray. What is the minimum total cost if you act optimally?

Input

The first input line has two integers n and k : the array elements and the number of subarrays. The array elements are numbered $1, 2, \dots, n$.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \leq k \leq n \leq 3000$
- $1 \leq x_i \leq 10^5$

Example

Input:

```
8 3
2 3 1 2 2 3 4 1
```

Output:

```
110
```

Explanation: An optimal solution is $[2, 3, 1]$, $[2, 2, 3]$, $[4, 1]$, whose cost is $(2 + 3 + 1)^2 + (2 + 2 + 3)^2 + (4 + 1)^2 = 110$.

CSES Problem Set

Houses and Schools

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n houses on a street, numbered $1, 2, \dots, n$. The distance of houses a and b is $|a - b|$. You know the number of children in each house.

Your task is to establish k schools in such a way that each school is in some house. Then, each child goes to the nearest school. What is the minimum total walking distance of the children if you act optimally?

Input

The first input line has two integers n and k : the number of houses and the number of schools. The houses are numbered $1, 2, \dots, n$.

After this, there are n integers $c_1, c_2, \dots, c_n$: the number of children in each house.

Output

Print the minimum total distance.

Constraints

- $1 \leq k \leq n \leq 3000$
- $1 \leq c_i \leq 10^9$

Example

Input:

```
6 2
2 7 1 4 6 4
```

Output:

```
11
```

Explanation: Houses 2 and 5 will have schools.

CSES Problem Set

Knuth Division

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n numbers, your task is to divide it into n subarrays, each of which has a single element.

On each move, you may choose any subarray and split it into two subarrays. The cost of such a move is the sum of values in the chosen subarray.

What is the minimum total cost if you act optimally?

Input

The first input line has an integer n : the array size. The array elements are numbered $1, 2, \dots, n$.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \leq n \leq 5000$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
5
2 7 3 2 5
```

Output:

```
43
```

CSES Problem Set

Apples and Bananas

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n apples and m bananas, and each of them has an integer weight between $1 \dots k$. Your task is to calculate, for each weight w between $2 \dots 2k$, the number of ways we can choose an apple and a banana whose combined weight is w .

Input

The first input line contains three integers k , n and m : the number k , the number of apples and the number of bananas.

The next line contains n integers $a_1, a_2, \dots, a_n$: weight of each apple.

The last line contains m integers $b_1, b_2, \dots, b_m$: weight of each banana.

Output

For each integer w between $2 \dots 2k$ print the number of ways to choose an apple and a banana whose combined weight is w .

Constraints

- $1 \leq k, n, m \leq 2 \cdot 10^5$
- $1 \leq a_i \leq k$
- $1 \leq b_i \leq k$

Example

Input:

```
5 3 4
5 2 5
4 3 2 3
```

Output:

```
0 0 1 2 1 2 4 2 0
```

Explanation: For example for $w = 8$ there are 4 different ways: we can pick an apple of weight 5 in two different ways and a banana of weight 3 in two different ways.

CSES Problem Set

One Bit Positions

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a binary string of length n . Your task is to calculate, for every k between $1 \dots n - 1$, the number of ways we can choose two positions i and j such that $i - j = k$ and there is a one-bit at both positions.

Input

The only input line has a string that consists only of characters 0 and 1.

Output

For every distance k between $1 \dots n - 1$ print the number of ways we can choose two such positions.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$

Example

Input:

1001011010

Output:

1 2 3 0 2 1 0 1 0

CSES Problem Set**Signal Processing**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given two integer sequences: a signal and a mask. Your task is to process the signal by moving the mask through the signal from left to right. At each mask position calculate the sum of products of aligned signal and mask values in the part where the signal and the mask overlap.

Input

The first input line consists of two integers n and m : the length of the signal and the length of the mask.

The next line consists of n integers $a_1, a_2, \dots, a_n$ defining the signal.

The last line consists of m integers $b_1, b_2, \dots, b_m$ defining the mask.

Output

Print $n + m - 1$ integers: the sum of products of aligned values at each mask position from left to right.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a_i, b_i \leq 100$

Example

Input:

```
5 3
1 3 2 1 4
1 2 3
```

Output:

```
3 11 13 10 16 9 4
```

Explanation: For example, at the second mask position the sum of aligned products is $2 \cdot 1 + 3 \cdot 3 = 11$.

CSES Problem Set**New Roads Queries**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities in Byteland but no roads between them. However, each day, a new road will be built. There will be a total of m roads.

Your task is to process q queries of the form: "after how many days can we travel from city a to city b for the first time?"

Input

The first input line has three integers n , m and q : the number of cities, roads and queries. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the roads in the order they are built. Each line has two integers a and b : there will be a road between cities a and b .

Finally, there are q lines that describe the queries. Each line has two integers a and b : we want to travel from city a to city b .

Output

For each query, print the number of days, or -1 if it is never possible.

Constraints

- $1 \leq n, m, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 4 3
1 2
2 3
1 3
2 5
1 3
3 4
3 5
```

Output:

```
2
-1
4
```

CSES Problem Set

Dynamic Connectivity

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider an undirected graph that consists of n nodes and m edges. There are two types of events that can happen:

1. A new edge is created between nodes a and b .
2. An existing edge between nodes a and b is removed.

Your task is to report the number of components after every event.

Input

The first input line has three integers n , m and k : the number of nodes, edges and events.

After this there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b . There is at most one edge between any pair of nodes.

Then there are k lines describing the events. Each line has the form " $t a b$ " where t is 1 (create a new edge) or 2 (remove an edge). A new edge is always created between two nodes that do not already have an edge between them, and only existing edges can get removed.

Output

Print $k + 1$ integers: first the number of components before the first event, and after this the new number of components after each event.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m, k \leq 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3 3
1 4
2 3
3 5
1 2 5
2 3 5
1 1 2
```

Output:

```
2 2 2 1
```

CSES Problem Set

Parcel Delivery

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m routes through which parcels can be carried from one city to another city. For each route, you know the maximum number of parcels and the cost of a single parcel.

You want to send k parcels from Syrjälä to Lehmälä. What is the cheapest way to do that?

Input

The first input line has three integers n , m and k : the number of cities, routes and parcels. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä and city n is Lehmälä.

After this, there are m lines that describe the routes. Each line has four integers a , b , r and c : there is a route from city a to city b , at most r parcels can be carried through the route, and the cost of each parcel is c .

Output

Print one integer: the minimum total cost or -1 if there are no solutions.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq k \leq 100$
- $1 \leq a, b \leq n$
- $1 \leq r, c \leq 1000$

Example

Input:

```
4 5 3
1 2 5 100
1 3 10 50
1 4 7 500
2 4 8 350
3 4 2 100
```

Output:

```
750
```

Explanation: One parcel is delivered through route $1 \rightarrow 2 \rightarrow 4$ (cost $1 \cdot 450 = 450$) and two parcels are delivered through route $1 \rightarrow 3 \rightarrow 4$ (cost $2 \cdot 150 = 300$).

CSES Problem Set

Task Assignment

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A company has n employees and there are n tasks that need to be done. We know for each employee the cost of carrying out each task. Every employee should be assigned to exactly one task. What is the minimum total cost if we assign the tasks optimally and how could they be assigned?

Input

The first input line has one integer n : the number of employees and the number of tasks that need to be done.

After this, there are n lines each consisting of n integers. The i th line consists of integers $c_{i1}, c_{i2}, \dots, c_{in}$: the cost of each task when it is assigned to the i th employee.

Output

First print the minimum total cost.

Then print n lines each consisting of two integers a and b : you assign the b th task to the a th employee.

If there are multiple solutions you can print any of them.

Constraints

- $1 \leq n \leq 200$
- $1 \leq c_{ij} \leq 1000$

Example

Input:

```
4
17 8 16 9
7 15 12 19
6 9 10 11
14 7 13 10
```

Output:

```
33
1 4
2 1
3 3
4 2
```

Explanation: The minimum total cost is 33. We can reach this by assigning employee 1 task 4, employee 2 task 1, employee 3 task 3 and employee 4 task 2. This will cost $9 + 7 + 10 + 7 = 33$.

CSES Problem Set

Distinct Routes II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A game consists of n rooms and m teleporters. At the beginning of each day, you start in room 1 and you have to reach room n .

You can use each teleporter at most once during the game. You want to play the game for exactly k days. Every time you use any teleporter you have to pay one coin. What is the minimum number of coins you have to pay during k days if you play optimally?

Input

The first input line has three integers n , m and k : the number of rooms, the number of teleporters and the number of days you play the game. The rooms are numbered $1, 2, \dots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b : there is a teleporter from room a to room b .

There are no two teleporters whose starting and ending room are the same.

Output

First print one integer: the minimum number of coins you have to pay if you play optimally. Then, print k route descriptions according to the example. You can print any valid solution.

If it is not possible to play the game for k days, print only -1.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq k \leq n - 1$
- $1 \leq a, b \leq n$

Example

Input:

```
8 10 2
1 2
1 3
2 5
2 4
3 5
3 6
4 8
5 8
6 7
7 8
```

Output:

```
6
4
1 2 4 8
4
1 3 5 8
```

